Lab-1

Program-1 :To print the hello world

In [1]:
```
print("hello world")
```

```
hello world
```

program2: to print sum of two numbers

In [2]:
```
a=10
b=2
c=a+b
#it will print the value of sum of a and b
print(c)
```

```
12
```

program3: to print square root

In [3]:
```
a=int(input('enter the number'))
sr=a**0.5
# sr will print the value of squareroot of value entered
print('square root of {0} is {1} '.format(a,sr))
```

```
enter the number4
square root of 4 is 2.0
```

program4: to print average of three numbers

In [4]:
```
a=int(input('enter the number '))
b=int(input("enter the number"))
c=int(input('enter the number'))
# d is taken as variable to print avg of 3 values entered
d=(a+b+c)/3
print(d)
```

```
enter the number 2
enter the number2
enter the number2
2.0
```

program5: find area of circle

In [5]:
```
a=float(input('enter the radius of circle'))
pi=3.14
# pi is fixed variable with the value of 3.14
# area is variable that is used to calculate the area of radius entered
area=pi*(a**2)
print(area)
```

```
enter the radius of circle10
314.0
```

program6: wap to swap two numbers

In [6]:
```
a=int(input('enter the number'))
b=int(input('enter the number'))
#to print the initial value of a and b
print('a={0} b={1}'.format(a,b))
a=a+b
b=a-b
a=a-b
#process involved in swapping the number without using 3rd variable
print('after swapping')
print('a={0} b={1}'.format(a,b))
```

```
enter the number1
enter the number2
a=1 b=2
after swapping
a=2 b=1
```

program7: find the root of qaudratic equation

In [1]:
```
import cmath
#to import math function from library
a=int(input('enter first coefficient'))
b=int(input('enter first coefficient'))
c=int(input('enter first coefficient'))
#to print roots of the equation of the values entered
d=(b**2)-(4*a*c)
a1=(-b - cmath.sqrt(d))/(2*a)
a2=(-b - cmath.sqrt(d))/(2*a)
print('the solution are :')
print('First root :',a1)
print('Second root :',a2)
```

```
enter first coefficient1
enter first coefficient2
enter first coefficient4
the solution are :
First root : (-1-1.7320508075688772j)
Second root : (-1-1.7320508075688772j)
```

example1: wap to convert temperature in cel to farenheit

In [2]:
```
c=float(input('enter the temperature in celsius'))
#f is declared as variable to calculate celsius to farenheit
f=1.8*c + 32
print('temperature in farenheit %d'%f)
```

```
enter the temperature in celsius40
temperature in farenheit 104
```

example2(a): wap find the solution

In [3]:
```
a=1
b=2
c=3
#a,b,c are already declared variable which is now used to calculate root
d=(b**2)-(4*a*c)
#inbuilt squareroot function is used from imported math function
a1=(-b - cmath.sqrt(d))/(2*a)
a2=(-b - cmath.sqrt(d))/(2*a)
print('the solution are :')
print(a1)
print(a2)
```

```
the solution are :
(-1-1.4142135623730951j)
(-1-1.4142135623730951j)
```

example2(b): wap to find solution of eaqutiona=(b+c)*(b-c)

In [4]:
```
b=10
c=15
#a is used to solve the equation given in question
a=(b+c)*(b-c)
print('solution of eqaution is')
print(a)
```

```
solution of eqaution is
-125
```

example2(c):wap to find solution of a=-(r1/r2+r3)

In [5]:
```
r1=int(input('enter the number'))
r2=int(input('enter the number'))
r3=int(input('enter the number'))
#r1,r2,r3 are user defined variable
#a is used to solve the equation given in question
a=-(r1/r2+r3)
print('solution of eqaution is')
print(a)
```

```
enter the number250
enter the number25
enter the number25
solution of eqaution is
-35.0
```

example2(d): wap to find solution of i= (p*r*t)/100

In [6]:
```
p=float(input('enter the price'))
r=float(input('enter the rate'))
t=float(input('enter the time'))
#user enters the value of price,rate of intrest and time
#i is used as a variable to calculate simple interest
i=(p*r*t)/100
print('the simple interest is')
print(i)
```

```
enter the price1000
enter the rate10
enter the time5
the simple interest is
500.0
```

example3:wap to accept base and height and area of a triangle

In [7]:
```
base=float(input('enter the base'))
#user enters the value of base of triangle
height=float(input('enter the height'))
#user enters the value of height of triangle
area=0.5*base*height
#area of triangle is calculated and printed
print('area of triangle is',area)
```

```
enter the base20
enter the height5
area of triangle is 50.0
```

example4: wap to find midpoint of line

In [8]:
```
x1=int(input('enter the x1 coordinates'))
y1=int(input('enter the y1 coordinates'))
x2=int(input('enter the x2 coordinates'))
y2=int(input('enter the y2 coordinates'))
#x1,x2 and y1,y2 are the coordinates of the line entered by user
m1=(x1+x2)/2
m2=(y1+y2)/2
#m1 and m2 are used to calculate the midpoint of line
print('midpont of line is({0},{1})'.format(m1,m2))
```

```
enter the x1 coordinates22
enter the y1 coordinates2
enter the x2 coordinates4
enter the y2 coordinates4
midpont of line is(13.0,3.0)
```

Example 5:wap a accept three number and print sum of cube of three number

In [9]:
```
a=float(input('Enter the number:'))
b=float(input('Enter the number:'))
c=float(input('Enter the number:'))
#a,b,c are user defined variable
#sum is used as var to calculate sum of cube of 3 numbers entered
sum= (a**3)+(b**3)+(c**3)
print('Sum of cube of three number is:')
print(sum)
```

```
Enter the number:12
Enter the number:2
Enter the number:5
Sum of cube of three number is:
1861.0
```

Example 6:wap to calculate number of seconds in a day

In [10]:
```
Day=int(input('Enter the day:'))
# to convert days into seconds
second=Day*24*3600
print('Seconds :',second)
```

```
Enter the day:5
Seconds : 432000
```

Example 7:wap to convert kilometer to miles

In [11]:
```
km=float(input('Enter the  distance kilometer:'))
# to convert km into miles
miles=km*0.621371192
# to print distance in miles
print('The distance in miles is:')
print(miles)
```

```
Enter the  distance kilometer:12
The distance in miles is:
7.456454304
```

Example 8:wao to accept two integer x and y from user abd then display the out

In [12]:
```
x=float(input('Enter the value of x:'))
y=float(input('Enter the value of Y:'))
z=x**y
# to print x raise to y entered by user
print("The x to the power of y is:")
print(z)
```

```
Enter the value of x:10
Enter the value of Y:2
The x to the power of y is:
100.0
```